

Homework 5

Fitting probability distributions to data

- A real number X is drawn from the Gaussian distribution $N(10, 16)$.
 - What is the probability that $X \geq 10$?
 - What is the probability that $X = 10$?
 - What is the probability (roughly) that $X \geq 14$?
 - What is the probability (roughly) that $X \leq 2$?
- A call center keeps track of the number of phone calls they receive: over a period of 500 hours, they record the number of calls received during every one-hour interval (the number of calls during the first hour, during the second hour, and so on). Let N_k be the number of one-hour intervals during which k calls were received, for $k = 0, 1, 2, \dots$. Here is their data:

k	0	1	2	3	4	5	6	7	8	≥ 9
N_k	22	66	106	115	85	55	28	13	10	0

Notice that $N_0 + N_1 + \dots = 500$.

- You decide to model the number of calls received in an hour by a $\text{Poisson}(\lambda)$ distribution. What value of λ should you choose?
 - Plot this distribution, showing the probabilities it assigns to outcomes $k = 0, 1, \dots, 10$. One way to do this is using `scipy.stats.poisson.pmf` to generate the probabilities and `matplotlib.pyplot.bar` to display them.
 - Under this choice of λ , what are the expected entries in the table above, i.e. the expected number of one-hour intervals (out of 500) during which k calls are received, for $k = 0, 1, \dots$? (The expected value for N_k is 500 times the probability of outcome k under $\text{Poisson}(\lambda)$.)
- Fitting a uniform distribution.* For any real number $\lambda > 0$, let U_λ denote the uniform distribution over $[0, \lambda]$.
 - Write down the density of U_λ , i.e., give a formula for the density at any point x .
 - Given a set of observations $x_1, x_2, \dots, x_n > 0$, we decide to fit a U_λ distribution to them. What is the maximum-likelihood choice of λ ? Just give the answer that seems intuitively correct to you; there is no calculus required.
 - Chi-squared distribution.* Suppose $Z_1, \dots, Z_k$ are independent $N(0, 1)$ random variables. Define $X = Z_1^2 + \dots + Z_k^2$. The distribution of X is called the chi-squared distribution with k degrees of freedom.
 - Plot the density of the chi-squared distribution with 10 degrees of freedom using `scipy.stats.chi2.pdf`.

- (b) By generating 10,000 random samples from this distribution (using `numpy.random.chisquared`), estimate its median.
5. *Maximum likelihood and smoothing.* Upon tossing a coin 20 times, you get heads every time.
- (a) How would you estimate the bias (that is, the heads probability) of the coin, using maximum likelihood?
 - (b) How would you estimate the bias of the coin, using maximum likelihood *with Laplace smoothing*?
 - (c) Under your estimate from part (b), what is the probability of seeing the sequence of tosses *HHTTHH*? You don't need to simplify the numeric expression.
6. *Fitting a multinomial.* Fix the following vocabulary: $V = \{\text{a, rose, is, flower}\}$.
- (a) In a bag-of-words representation, what is the vector form of the document “A rose is a rose is a rose”?
 - (b) Fit a multinomial model to this document using maximum likelihood but not Laplace smoothing. What is the resulting distribution (give the probability of each word in V)?
 - (c) Same as part (b), but with Laplace smoothing.
7. *Text generation using n -grams.* In this question, we will try to emulate the language of Shakespeare using an n -gram language model. We will base the model on a collection of Shakespeare's sonnets and plays: the text and an accompanying Jupyter notebook are provided on the course webpage, in `text-gen.zip`.
- Take a careful look through the Jupyter notebook. In the implementation, `<START>` tokens are appended at the beginning of each sentence as indicators. A fullstop (period) token is used to indicate the end of a sentence.
- (a) Create n -gram models for $n = 2, 3$ and, for each of these values of n , generate a 100-word text using the trained model.
 - (b) Complete the function `model_ll`, which takes a sentence and an n -gram model, and returns the log likelihood of the sentence under that model. Use this function to get the log-likelihood of the provided sentence under unigram and bigram models.
 - (c) Given a prompt from one of Shakespeare's sonnets, generate the next 20 words using the bigram and trigram models, and report the two texts.

Submission format: For part (b), screenshot and crop your function implementation and assign it to the respective section on gradescope. Only the function code should be attached.

For parts (a) and (c), show the texts that are generated.

For part (b), give the log-likelihood of the sentence under the two models.

Bayesian inference

8. Let θ denote the probability that you win when playing tennis with your friend. You want to estimate this probability, and you decide to try both *frequentist* and *Bayesian* approaches.
- You play ten matches with your friend. The outcomes are:

$$X = (W, W, L, L, L, L, L, W, W, W)$$

where W means that you win and L means that you lose.

- (a) Let's begin with the frequentist approach, in which you estimate the *maximum-likelihood* value of θ . What is this value, given the observations above?
 - (b) Now let's follow a Bayesian approach. This time, we need to begin with a *prior distribution* on the quantity of interest, θ . Based on your prior knowledge about your relative tennis abilities, you put a $\text{Beta}(6,4)$ prior distribution on θ .
What is $\mathbb{E}[\theta]$, the mean of the prior distribution? What is its standard deviation, $\text{std}(\theta)$?
 - (c) Now let's continue with the Bayesian approach by figuring out the *posterior distribution* of θ given observations X . What is this posterior distribution? What is $\mathbb{E}[\theta|X]$, the mean of the posterior given X ? What is the standard deviation of the posterior?
 - (d) Notice how the Bayesian approach gives us a *distribution* over possible values of θ , rather than a single choice of θ . Plot the density of this posterior distribution.
9. *Accidents per day*. The number of car accidents per day in a big city follows a $\text{Poisson}(\theta)$ distribution. Suppose that we decide to do a Bayesian analysis in which we treat θ as a random variable whose prior distribution is $\text{gamma}(1,1)$.
- (a) Under this prior, what is the expected value of θ ?
 - (b) Now we get some data. We count the number of car accidents for ten days, and get the following values:
$$13, 2, 4, 20, 10, 9, 4, 5, 12, 15.$$
- Based on these observations, what is the *posterior* distribution of θ ?
- (c) Continuing from part (b), plot both the prior and posterior densities.
10. In an election, there are three candidates, A , B , and C . Of the voting population, θ_A fraction will vote for A , while θ_B fraction will vote for B and θ_C fraction will vote for C ; thus $\theta_A + \theta_B + \theta_C = 1$.
- (a) John wishes to infer the distribution $(\theta_A, \theta_B, \theta_C)$. He starts by putting a uniform prior on it; that is, he considers all distributions to be equally likely. He then picks four people at random to poll, and finds that they intend to vote for A, B, B, A , respectively. Given this information, what is John's posterior distribution?
 - (b) Under this posterior distribution, what is the expected value of $(\theta_A, \theta_B, \theta_C)$?